

UEC513: EMBEDDED SYSTEMS				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: The objective of this course is to equip students with the necessary fundamental knowledge and skills that enable them to design basic embedded systems. It covers architecture, programming of ARM processor and its interfacing with peripheral devices.				
Syllabus Introduction to Embedded Systems: Definition, Embedded Systems Vs General Computing Systems, Classification of Embedded Systems, Major application areas. General purpose processor architecture and organization, Von-Neumann and Harvard architectures, CISC and RISC architectures, Big- and Little-endian processors, Processor design trade-offs, Processor cores: soft and hard. Introduction to ARM Processor: The ARM design philosophy, ARM core data flow model, Architecture, Register set, ARM7TDMI Interface signals, General Purpose Input Output Registers, Memory Interface, Bus Cycle types, Operational Modes, Pipeline: ARM 3 stage Pipeline, ARM family attribute comparison. ARM 5 stage Pipeline, Pipeline Hazards, Data forwarding. Programming based on ARM7TDMI: ARM Instruction set, condition codes, Addressing modes, Interrupts, Exceptions and Vector Table. Instruction Format Assembly Language Programming, Thumb state, Thumb Programmers model, Thumb Instruction format, Thumb Applications, ARM coprocessor interface and Instructions. ARM Tools and Interfacing of Peripherals: ARM Development Environment, Arm Procedure Call Standard (APCS), Example C/C++ programs, Embedded software development, Image structure, linker inputs and outputs, Protocols (I2C, SPI), Memory Protection Unit (MPU). Physical Vs Virtual Memory, Paging, Segmentation. The Advanced Microcontroller Bus Architecture (AMBA), DMA, Peripherals, Interfacing of peripherals with ARM. Familiarization with Standards: IEEE 1275.1-1994 and IEEE 1754.				
Laboratory Work: Introduction to Kiel Software, Introduction to ARM processor kit, Programming examples of ARM processor. Interfacing of LED, Seven Segment Display, Stepper Motor, LCD with ARM7TDMI processor.				
Course Learning Objectives (CLO) The students will be able to: 1. Explain embedded systems, their processor architecture, and distinguish them from general-purpose computing systems. 2. Describe the internal architecture of ARM processors, their instructions and formats, and create ARM-based programs for specific tasks. 3. Interpret thumb mode operations and integrate various coprocessors in embedded systems. 4. Analyze different pipeline structures and evaluate hazards. 5. Demonstrate the interfacing of various hardware peripherals in embedded systems.				
Text Books 1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Naraig Manjikian, "COMPUTER				

- | |
|---|
| <p><i>ORGANIZATION AND EMBEDDED SYSTEMS</i>, Sixth Edition, McGraw Hill, 2012.</p> <p>2. <i>Steve Furber, "ARM System-on-Chip Architecture, Second Edition, PEARSON, 2013.</i></p> |
| <p>Reference Books</p> <p>1. <i>Stephen Welsh, Peter Knaggs, "ARM: Assembly Language Programming", Bourne Mouth University Publication, 2003.</i></p> <p>2. <i>Andrew N. Sloss, Dominic Symes, Chris Wright "ARM System Developers Guide, Designing and Optimizing System Software", Elsevier Publication.</i></p> |